

Weekly Problem 5

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Exercise 1

Part (a)

1	$(p \vee q) \Rightarrow r$	
2	$\neg(p \vee q) \vee r$	mcE1
3	$\neg(p \vee q)$	
4	$\neg p$	$\neg\vee E3$
5	$\neg\neg r \vee \neg p$	$\vee I4$
6	r	
7	$\neg\neg r$	$\neg\neg I6$
8	$\neg\neg r \vee \neg p$	$\vee I7$
9	$\neg\neg r \vee \neg p$	$\vee E2, 3-5, 6-8$
10	$\neg r \Rightarrow \neg p$	mcI9

We then see that $(p \vee q) \Rightarrow r \vdash \neg r \Rightarrow \neg p$ is a valid argument in the Logic of First-Degree Entailment.

The derivation only used rules that were available in both Strong Kleene logic and the Logic of Paradox, so the argument is valid in those logics as well.

Part (b)

Let $p = b$ and $q = n$. We then see that $\neg p = b$, so $\neg p \Rightarrow q = 1$ and $\neg p \Rightarrow q = 1$. However, $q \Rightarrow q = n$. Consequently, the argument $p, (p \wedge q) \Rightarrow r \vdash q \Rightarrow r$ is an invalid argument in the Logic of First-Degree Entailment.

However, this argument is valid in Strong Kleene logic and the Logic of

Paradox, as seen by the following derivation.

1	$\neg p \Rightarrow q$	
2	$p \Rightarrow q$	
3	$\neg\neg p \vee q$	mcE1
4	$\neg p \vee q$	mcE2
5	$\neg p$	
6	q	$\Rightarrow$ E1, 5
7	$\neg q \vee q$	$\vee$ I6
8	q	
9	$\neg q \vee q$	$\vee$ I8
10	$\neg q \vee q$	$\vee$ E3, 5–7, 8–9
11	$q \Rightarrow q$	mcI10

The argument only fails in the Logic of First-Degree Entailment because of the truth values b and n , which were not present in Strong Kleene logic or the Logic of Paradox.